Quiz — 1.3.3 Networks

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.3.3

Section A — Multiple choice (8 marks)

1. Which best describes a LAN? A. A network spanning cities or countries using third-party infrastructure B. A network over a small geographical area where the organisation owns the hardware C. A network with no physical hardware D. The largest WAN in the world

Your answer: ☐

2. In a peer-to-peer network: A. A central server provides all services B. All devices are equal and can act as client and server C. There is always a single point of failure D. Security is centrally managed

Your answer: ☐

3. A disadvantage of a star topology is that: A. A single cable fault brings down every device B. The central switch/hub is a single point of failure C. It cannot use a switch D. It requires every node to connect to every other node

Your answer: ☐

4. In packet switching, packets are: A. Sent along one dedicated reserved path B. Routed independently and reassembled at the destination C. Never reassembled D. Only used for telephone calls

Your answer: ☐

5. Which protocol is used to *send* email? A. POP3 B. IMAP C. SMTP D. FTP

Your answer: ☐

6. The DNS is responsible for: A. Encrypting web traffic B. Translating domain names into IP addresses C. Splitting data into packets D. Assigning MAC addresses

Your answer: ☐

7. Which device connects two different networks and routes using IP addresses? A. Switch B. Router C. NIC D. Wireless access point

Your answer: ☐

8. At which TCP/IP layer are source and destination IP addresses added? A. Application B. Transport C. Network/Internet D. Link/Data link

Your answer: ☐

Section B — Short answer (12 marks)

9. State two advantages of a client-server network over a peer-to-peer network. **[2]**

10. TCP/IP layer matching. Match each item (A–D) to the correct layer by writing the layer name next to it. **[4]**

A. Adds the MAC address / handles the physical connection → ________

B. Adds source and destination IP addresses and routes packets → ________

C. Splits data into packets, manages ports and end-to-end delivery → ________

D. Provides services for user applications (e.g. HTTP, SMTP) → ________

11. Explain the difference between a switch and a router. **[2]**

12. Describe what a proxy server does and give one benefit of using one. **[2]**

13. Explain one reason packet switching is more resilient than circuit switching. **[2]**

Section C — Exam-style question (5 marks)

14. A user types a web address into their browser and requests a secure web page over the internet.

Describe what happens, referring to: the role of DNS, how the data is broken up and routed across the network, and one piece of hardware involved. **[5]**